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PROFESSIONAL SUMMARY

AI Engineer and Medical Imaging Researcher specializing in deep learning, medical image segmentation, and bio-inspired optimization algorithms. Experienced in designing neural plasticity-inspired optimization frameworks for hyperparameter tuning and implementing U-Net/Attention U-Net architectures for brain MRI analysis. Skilled in end-to-end medical imaging pipelines using PyTorch and MONAI. Seeking graduate research opportunities in Computer Science focusing on computational intelligence, medical image analysis, and AI-driven healthcare systems.

PROFESSIONAL EXPERIENCE

Freelance Machine Learning Engineer 2023 - Present

Self-employed

- Developed machine learning models for classification and segmentation tasks in research environments.
- Designed hyperparameter optimization pipelines for GPU-based training (Google Colab, AWS EC2).
- Created technical documentation and research manuscripts using LaTeX.
- Supported academic research projects in deep learning and medical imaging.

EDUCATION

Master of Science in Computer Engineering (AI & Data Engineering) 2025

Bahir Dar University, Ethiopia | GPA: 3.45/4.0

Bachelor of Science in Electrical and Computer Engineering 2021

Debre Berhan University, Ethiopia

TECHNICAL SKILLS

Deep Learning & AI	PyTorch, TensorFlow, Keras, MONAI, Scikit-learn
Medical Imaging	NIFTI, DICOM, SimpleITK, NiBabel, OpenCV
Optimization & AI Methods	Bio-inspired Optimization, Neural Plasticity Learning, Genetic Algorithms, Particle Swarm Optimization (PSO), Hyperparameter Optimization
Data Engineering	NumPy, Pandas, SciPy, SQL, Apache Spark
MLOps & Tools	Linux, Docker, MLflow, Git, Jupyter Notebook, Google Colab, Overleaf
Programming Languages	Python, C++, Java, SQL, LaTeX
Cloud Platforms	AWS, Google Cloud Platform (GCP), Microsoft Azure

RESEARCH EXPERIENCE

Graduate Researcher - Master's Thesis

2023 - 2025

Bahir Dar University

Thesis: Reward-Driven Neural Plasticity Inspired Optimization for Enhancing U-net Based Medical Image Segmentation

- Designed a biologically inspired optimization algorithm based on neural plasticity and reward-driven learning.
- Developed deep learning pipelines using PyTorch and TensorFlow for brain MRI segmentation (tumor & stroke lesions).
- Implemented preprocessing workflows including normalization, resampling, and augmentation for NIfTI and DICOM datasets.
- Evaluated performance using Dice Coefficient, IoU, Precision, and Recall metrics.
- Achieved improved convergence stability and segmentation accuracy compared to Random Search and Genetic Algorithms.

Neuro-Inspired U-Net Optimization Project (Open Source)

Lead Developer | GitHub Project

- Developed a reproducible Python framework integrating bio-inspired optimization into deep learning training.
- Implemented Attention U-Net models for fine-grained medical image segmentation.
- Built automated medical imaging pipelines using MONAI and SimpleITK (skull stripping, preprocessing, artifact removal).
- Designed modular codebase for research reproducibility and extension.

RESEARCH INTERESTS

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| • Medical Image Analysis (MRI / CT segmentation, neuroimaging) | • Deep Learning Optimization and Architecture Design |
| • Bio-inspired and Neuro-inspired Learning Systems | • Domain Adaptation in Clinical AI |
| • Explainable AI (XAI) for Healthcare Systems | • Pediatric Neuroimaging and Tumor Analysis |